



Zearn Impact Study: 2024 Quasi-Experimental Study Across Four Colorado Districts

Zearn in Colorado

In the spring of 2023, Governor Polis and bipartisan legislators announced investments in data-driven math support for Colorado students, which included statewide access to a K-8 digital math accelerator, Zearn Math. In the school year 2023-24, Zearn partnered with CDE to roll out access to Zearn. In the first six months of the program, 920+ CO schools opted in to use Zearn Math for math acceleration, alongside teacher instruction.

For maximum impact, Zearn recommends students complete 3+ lessons-per-week, which is considered fidelity Zearn Math usage. This generally means 90 minutes-per-week on the platform. Zearn also looks at two other levels of consistent usage to capture the incremental effects of Zearn Math on student achievement. During 2023-24, 20K students in G1-8 used Zearn Math consistently, completing lessons each week of the school year, and 3.5K students in G1-8 used Zearn Math with fidelity (3+ lessons-per-week) across Colorado.

Zearn, in partnership with CDE, used a limited sample to understand Zearn Math's impact because of the lengthy process to collect data from each district individually. The study focused on a small subset of districts that represented different geographical and demographic areas of the state and had a large population of students with a wide range of usage.

Research questions

The Impact Study draws a sample from 6K G4-8 students who used Zearn Math consistently across the subset of Colorado districts, representing both urban and rural settings, during 2023-24 to answer two key research questions:

- 1 What is the effect on student achievement on Colorado Measures of Academic Success (CMAS) for G4-8 students who use Zearn Math consistently* compared to those who do not use Zearn Math?
- 2 What is the effect on student achievement on Colorado Measures of Academic Success (CMAS) for subgroups of students who use Zearn with fidelity (3+ lessons-per-week) compared to those who do not use Zearn Math?

Research methodology

This study assesses how Zearn Math usage in 2023-24 impacted math scores on the CMAS for G4-8 students who used Zearn consistently in a small set of CO districts representing diverse geographic and demographic areas of the state. Using a treatment-on-the-treated design, researchers compared students who completed Zearn lessons consistently (2.3K students), completing at least one lesson each week of the school year, to a matched group of non-users. The treatment group was segmented by usage levels: 1-1.9, 2-2.9, and 3+ grade-level lessons per week (fidelity usage).

The coarsened exact matching method was used to ensure comparable groups. Students were matched based on spring 2023 CMAS math and ELA scores, grade level, test language, and at least 6 of 9 additional demographic factors to isolate the impact of Zearn usage. Regression models were run on scale score data standardized around state means and standard deviations for each grade to account for grade-level differences in scaling in order to accurately compare results across grade-levels on the CMAS (a non-vertically scaled assessment). Results are presented as effect sizes, which quantify the magnitude of impact Zearn Math usage on students' CMAS performance for consistent Zearn users.

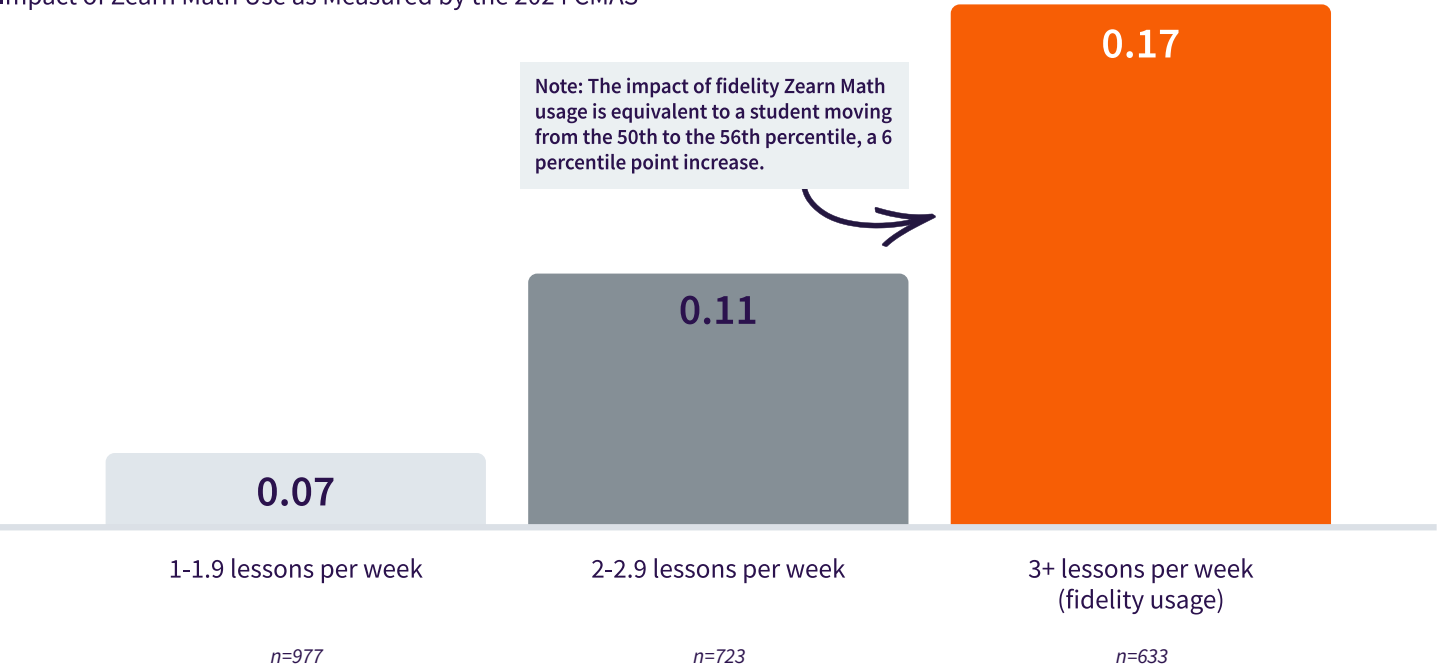
*Defined as completing at least one grade-level lesson per week during the academic school year.

Key findings

KEY FINDING 1

Consistent use of Zearn Math had a positive impact on achievement, as measured by CMAS math scores, with the highest impacts observed with fidelity usage (3+ grade-level lessons per week)

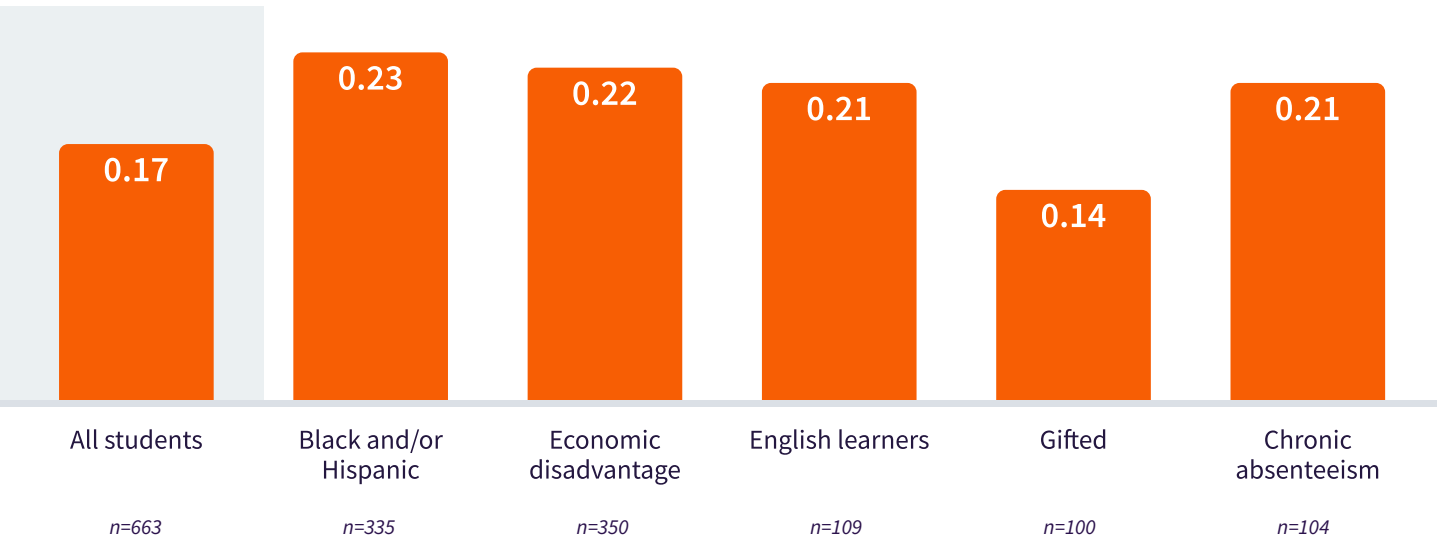
Impact of Zearn Math Use as Measured by the 2024 CMAS



KEY FINDING 2

Fidelity Zearn usage (3+ lessons-per-week) resulted in positive impacts in achievement across subgroups

Impact of Zearn Math on Student Achievement as Measured by CMAS 2024 by Subgroup

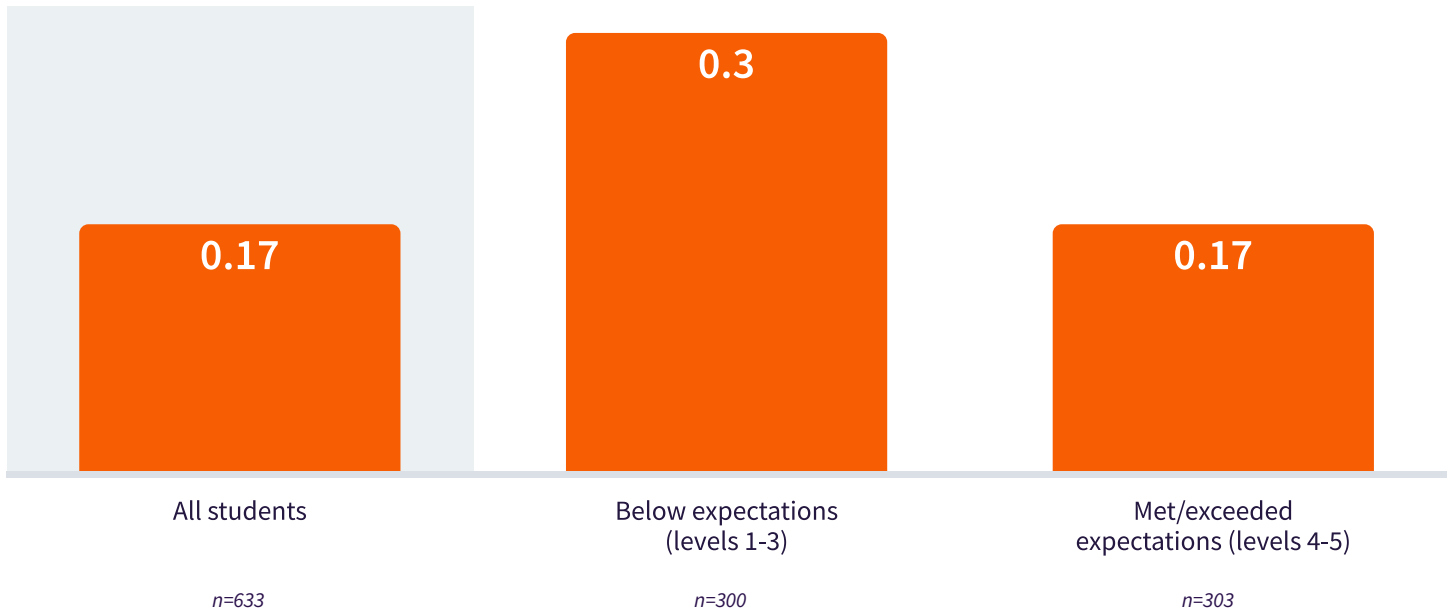


Effect sizes reported are Hedges' g. Subgroups with fewer than 100 students are excluded to ensure robustness of results. All results are statistically significant.

KEY FINDING 3

Fidelity Zearn usage (3+ lessons-per-week) resulted in positive impacts in achievement across prior-year CMAS Performance Levels

Impact of Zearn Math on Student Achievement as Measured by CMAS 2024 by Prior-Year Performance Level



Regression analysis detail

- Analysis is based on a regression model: $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \varepsilon$ in which Y represents 2024 OST math scores, while $\beta_1 \dots \beta_n$ represent student-level characteristics, including treatment or control, standardized 2023 CMAS math and ELA scores, gender, race, economic disadvantage, English learner, students with disabilities, gifted, and chronic absenteeism.
- Scale score data were standardized based on state means and standard deviations for each grade level to adjust for grade-level differences in scaling and allow comparison of results across grade-levels on CMAS (a non-vertically scaled assessment).
- One-level regression models are used because treatment and control students from the same school may be matched.
- Effect sizes reported are Hedges' g . Subgroups with fewer than 100 students are excluded to ensure robustness of results.
- Changes in percentile rankings are derived from regression coefficients (z-scores). The percentile ranks mentioned refer to a student's position within the sample, not the broader state population.
- More information on Zearn's methodology for treatment-on-the-treated impact analyses at webassets.zearn.org/Implementation/EfficacyResearchMethodology.pdf.

Effect sizes are Hedges' g . Percentiles based on regression coefficients. Subgroups with fewer than 100 students are excluded to ensure robustness. All results are statistically significant.

Zearn is the 501(c)(3) nonprofit educational organization behind Zearn Math, the top-rated math learning platform used by 1 in 4 elementary-school and 1 million middle-school students nationwide. Everything we do is driven by the belief that every kid is a math kid. Learn more at about.zearn.org.